

Speech Production and Models

Assignment 1 - Complete Results

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Assignment overview I

What was done

- Bandwidth and intelligibility study using low-pass filtering.
- Voiced and unvoiced analysis in time and frequency domains.
- LPC-based formant and spectral-envelope analysis.
- Manual phoneme labeling and spectrogram analysis.
- Frame-wise parameter estimation: energy, ZCR, voicing, LPC, and pitch.
- Progressive LPC vocoder from Q1 to Q5, ending with tuned WAV outputs.

Speech material used in the notebook

- High-rate signals for the bandwidth experiment: `male44.wav`, `female44.wav`
- Narrowband signals for analysis and synthesis: `male_short`, `male_long`, `female_long`
- Main analysis and synthesis sampling rate: **8 kHz**

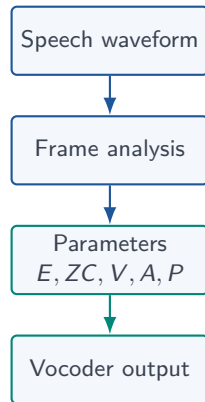
Assignment overview II

Main final settings

- Analysis length: **alen** = 256
- Update length: **ulen** = 80
- LPC order: **M** = 12
- Male pitch range: **70–250 Hz**
- Female pitch range: **90–350 Hz**

Presentation logic

- First: analysis results for each task.
- Then: parameter extraction results.
- Finally: Q1–Q5 vocoder progression and tuned outputs.



Bandwidth of speech: complete written results

Q1: At what cut-off does speech become intelligible?

Speech starts to become intelligible around **1 kHz**. Around **3 kHz** it becomes quite intelligible, because consonant cues and higher vowel formants start to be preserved.

Q2: At what cut-off does the quality start to degrade?

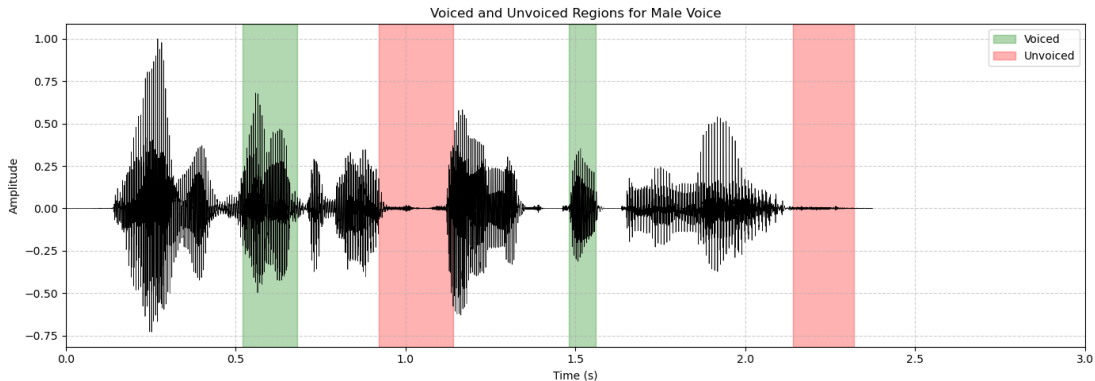
The sound starts to become muffled around **7–8 kHz**. Removing these higher frequencies suppresses fricative energy and upper harmonics, so the speech loses crispness.

Q3: Why is 8 kHz used in plain telephony?

The most important information for intelligibility lies roughly below **4 kHz**. By Nyquist, that suggests a sampling frequency of about **8 kHz**, which is a practical compromise between quality and transmission cost.

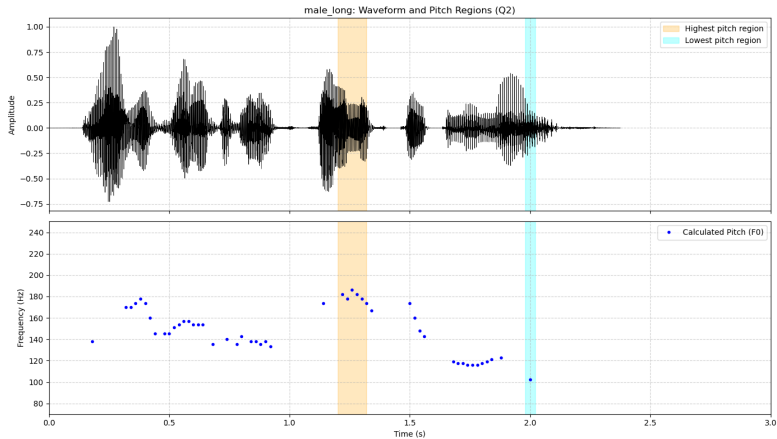
Takeaway: intelligibility survives with limited bandwidth, but naturalness depends strongly on upper-band speech cues.

Voiced and unvoiced regions on the waveform



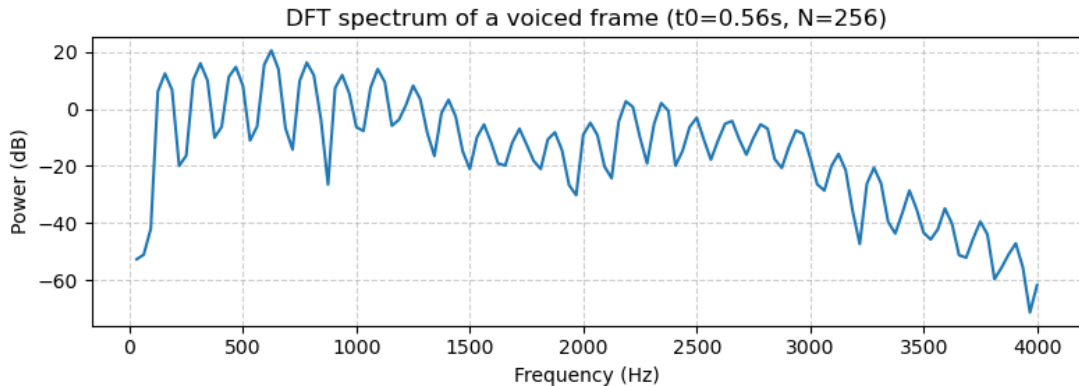
Manually marked voiced and unvoiced regions on `male_long`. Voiced regions show stronger periodicity and lower zero-crossing rate.

Highest and lowest pitch regions



Frame-wise pitch track with representative high-pitch and low-pitch intervals highlighted directly on the signal.

Voiced-frame spectrum



Q5

- Example voiced frame at about $t_0 = 0.56\text{ s}$ with clear harmonic structure.
- Estimated pitch for this frame: $f_0 \approx 157\text{ Hz}$ from about $L \approx 51$ **samples**.

Voiced-frame spectrum: written answers

Q4: What is the third FFT element?

$$X[k] = \sum_{n=0}^{N-1} x_f[n] e^{-j2\pi kn/N}$$

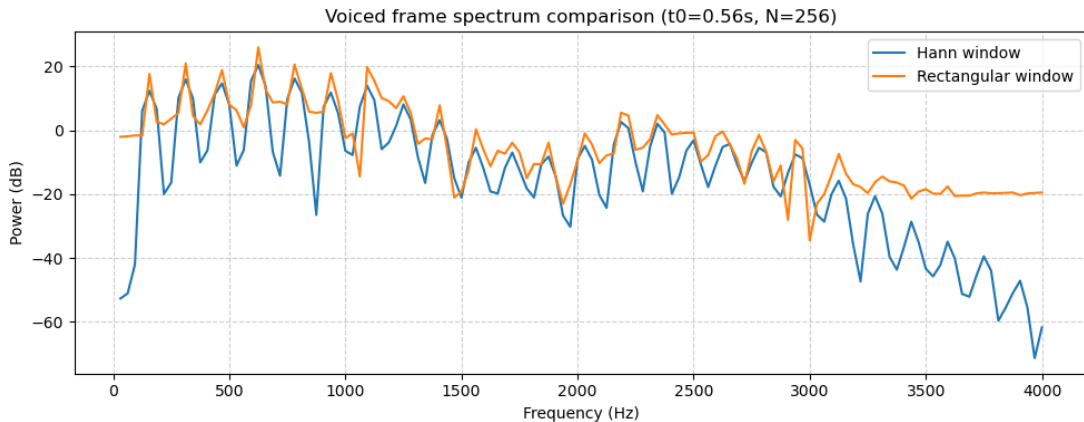
For $N = 256$ and $f_s = 8000$ Hz, the third element is $X[2]$ and it corresponds to

$$f_2 = \frac{2f_s}{N} = \frac{2 \cdot 8000}{256} = 62.5 \text{ Hz.}$$

Q6

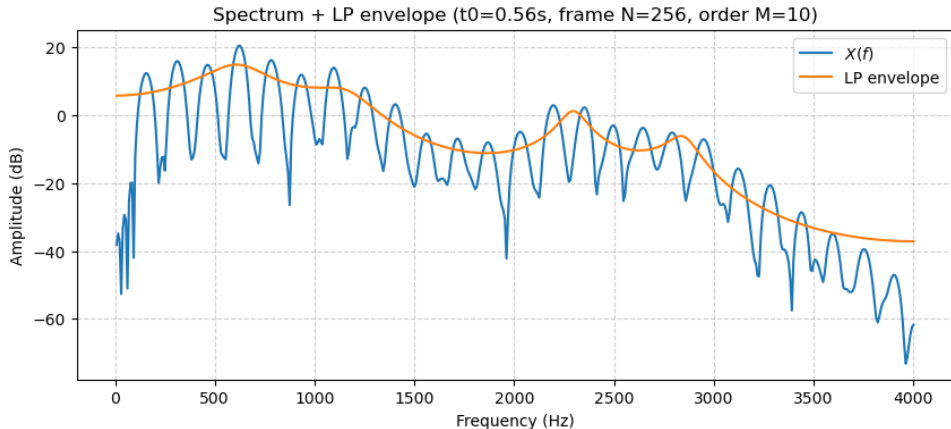
- A practical voiced-frame duration is about **20–40 ms**.
- Longer frames improve frequency resolution, while shorter frames improve time tracking.

Window comparison for the same voiced frame



The rectangular window has higher sidelobes and more leakage, while the Hann window gives a cleaner view of the harmonic structure.

LPC spectral envelope on the voiced frame



Q10

- The LPC envelope follows the broad spectral shape rather than the individual harmonics.
- For narrowband 8 kHz speech, a typical useful LPC order is about $M = 10$ to 12; the final system uses $M = 12$.

Autocorrelation and LPC scaling: written answer

Q9a: third element returned by `xcorr`

For nonnegative lags, the cross-correlation is

$$r_{xy}[k] = \sum_{n=0}^{N-1-k} x[n+k]y[n].$$

The third returned value corresponds to lag $k = 2$, so

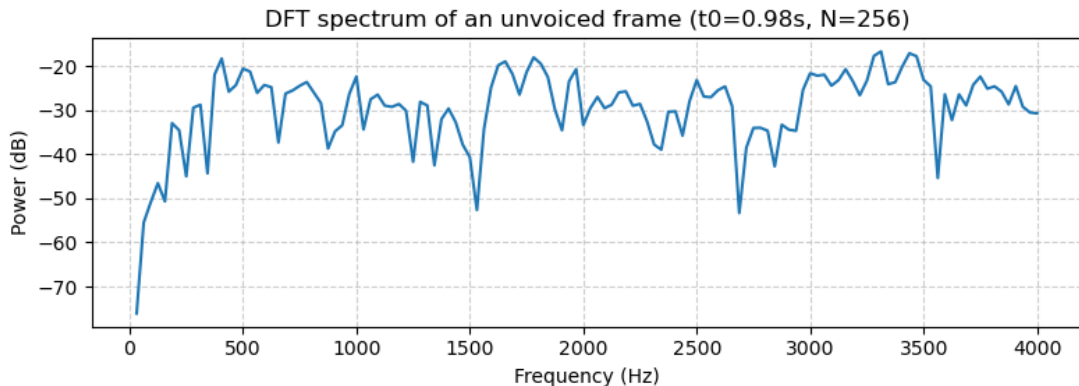
$$r_{xy}[2] = \sum_{n=0}^{N-3} x[n+2]y[n].$$

For autocorrelation of a frame $x_f[n]$, this becomes $r_{x_fx_f}[2] = \sum_{n=0}^{N-3} x_f[n+2]x_f[n]$.

Q9b: why multiply $|H(e^{j\omega})|^2$ by e ?

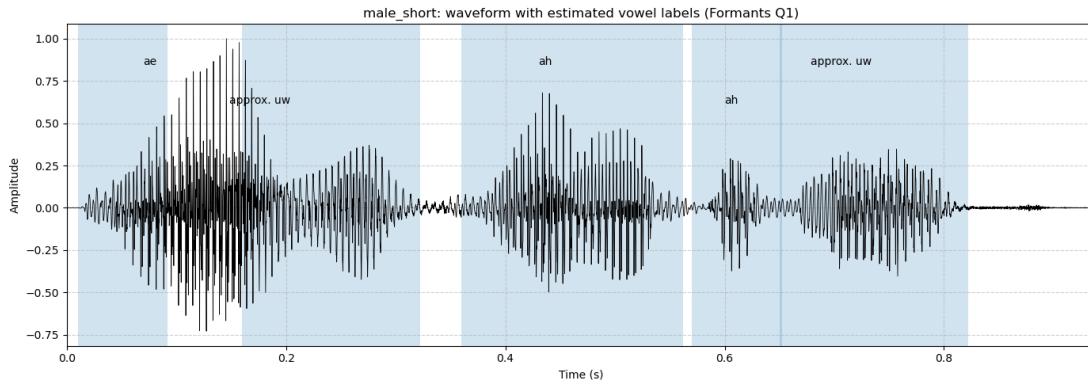
Under the AR model, the speech-frame power spectrum is $S_x(\omega) = e |H(e^{j\omega})|^2$, where e is the prediction-error variance. Multiplying by e puts the LPC envelope on the correct power scale.

Unvoiced-frame spectrum



- The unvoiced frame does not show a clean harmonic comb.
- Energy is spread more irregularly across frequency, which is consistent with noise-like excitation.

Formants: vowel labeling from LPC analysis



Estimated vowel labels in the notebook include **ae**, **approx. uw**, and **ah**; short transition regions are marked as approximate.

Formants: written interpretation

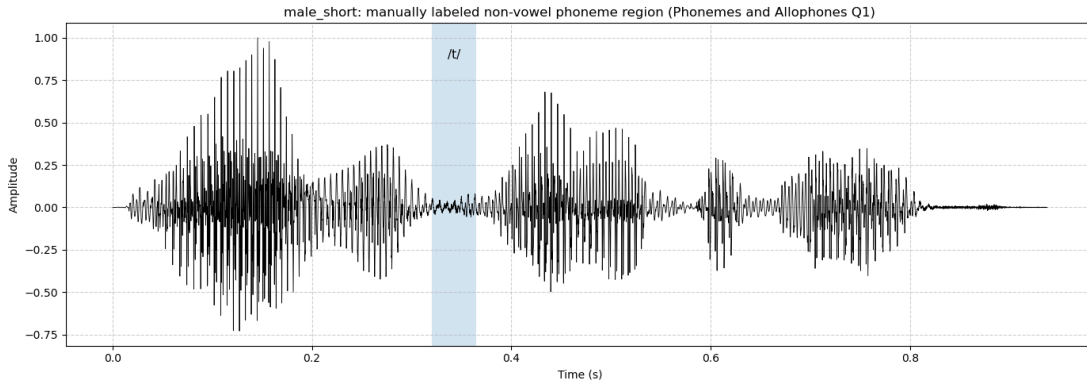
Q2: What is the relation between pitch and formants?

Pitch F_0 belongs to the excitation source and controls the spacing of the harmonic lines. Formants $F_1, F_2, F_3, \dots$ belong to the vocal-tract filter and determine the smooth spectral envelope.

Implication for the analysis

Changing F_0 moves the harmonic lines, but it does not move the formant resonances themselves. That is why pitch and formants must be interpreted as different physical parts of the speech-production model.

Phonemes and allophones: manual labeling result



Example manual non-vowel labeling on `male_short`; the notebook explicitly labels a */t/* region.

Phonemes and allophones: written theory answers

Q2: Can consonants have formant frequencies?

Consonants can show formant-like resonances in some cases, especially sonorants such as nasals, liquids, and approximants. Fricatives and stops are characterized differently.

Q3 and Q4

- A diphthong is a single vowel segment with changing vowel quality and moving formants.
- English is often described as having about **40–44 phonemes**, depending on dialect.

Q5: What is a phone? What is an allophone? How many allophones are there?

A **phone** is any concrete speech sound. An **allophone** is a context-dependent realization of a phoneme. There is no single fixed number of allophones, because the count depends on how finely the language is analyzed phonetically.

Spectrogram implementation: frame extraction view

Q1: Frame extraction principle

The function extracts overlapping frames from the speech signal. The analysis length determines the frame duration, while the update length determines the shift between consecutive frames.

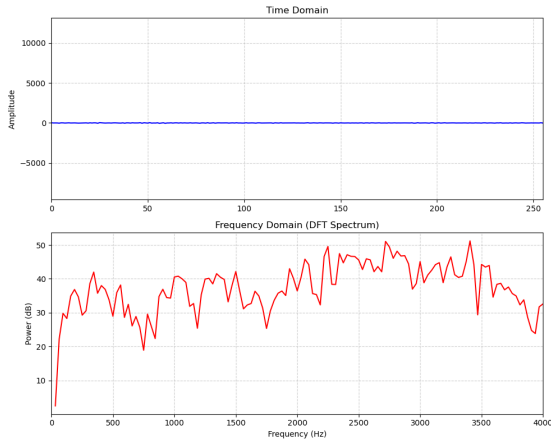
Interactive notebook visualization

This part of the implementation is interactive in the notebook. The visualization updates the selected frame position and displays the corresponding short-time segment of the waveform.

Interpretation

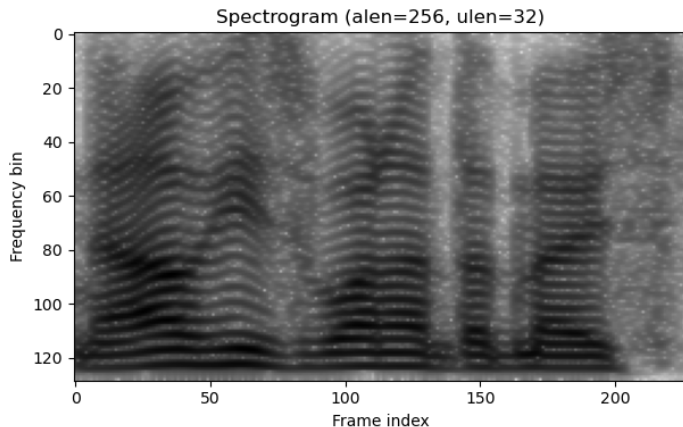
The frame-extraction view illustrates how a continuous speech signal is converted into a sequence of overlapping short-time frames for subsequent spectral analysis.

Spectrogram implementation: frame and DFT view



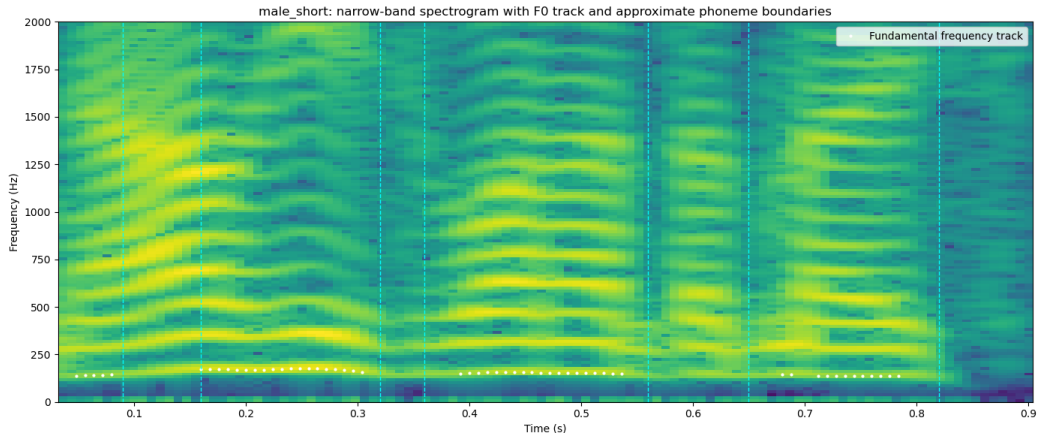
Q2 result: time-domain frame together with its DFT spectrum, illustrating the link between local waveform content and short-time frequency analysis.

Classical spectrogram



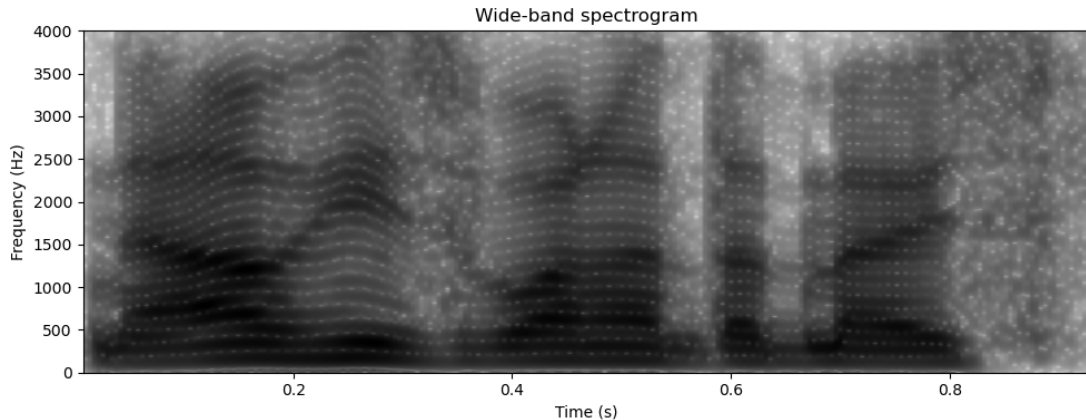
Q3 result: classical spectrogram produced by overlapping windowed DFT frames and assembling the magnitude columns over time.

Narrow-band spectrogram



- Narrow-band analysis makes the harmonic stripes clearer.
- This representation makes the pitch evolution easier to observe.

Wide-band spectrogram



- Wide-band analysis gives better time resolution.
- Rapid transitions and segment boundaries become easier to see.

Spectrogram comparison: narrow-band versus wide-band

Notebook conclusion

The wide-band spectrogram provides better time resolution, so rapid transitions and segment boundaries become clearer. The narrow-band spectrogram provides better frequency resolution, so harmonic structure and pitch are easier to observe.

Interpretation

Wide-band views are better for timing and transitions, while narrow-band views are better for periodicity and harmonic analysis.

Speech parameter estimation I: extracted parameters

Frame-wise parameters extracted in the notebook

$$E[n] = \frac{1}{N} \sum_m x_n^2[m]$$

$ZC[n]$ = fraction of sign changes in frame n

$V[n] \leftarrow$ decision from $E[n]$ and $ZC[n]$

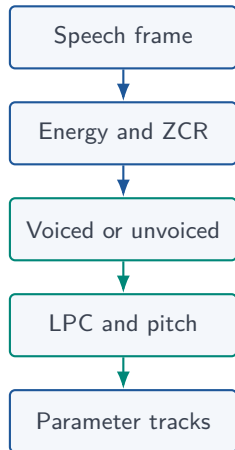
$\underline{a}_n$: LPC coefficients from frame autocorrelation

$P[n]$: pitch period from ACF peak in voiced frames

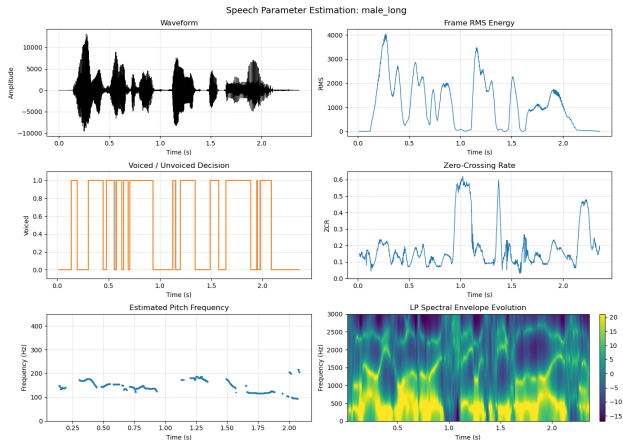
Speech parameter estimation II: analysis pipeline and design choices

Design choices in the code

- Hann analysis window.
- Toeplitz solution of the LPC normal equations.
- Pitch search restricted to a valid lag range.
- Smoothed voicing mask and smoothed pitch track in the tuned version.

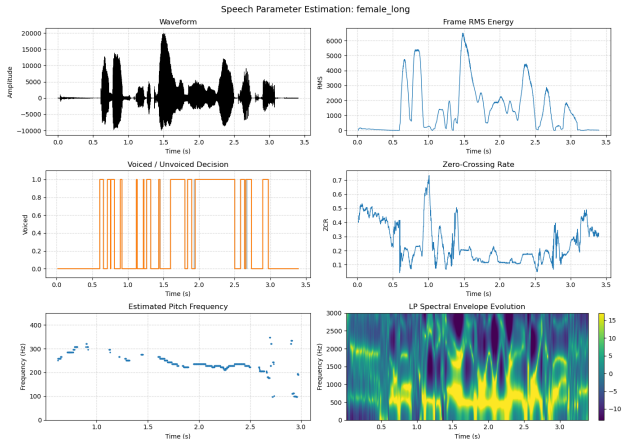


Speech parameter estimation III: male long signal



Male long utterance: waveform, energy, voicing, ZCR, pitch, and LPC spectral evolution.

Speech parameter estimation IV: female long signal



Female long utterance: same analysis pipeline, with a higher expected pitch range than the male signal.

Speech parameter estimation V: written discussion

Q6: What can make voiced or unvoiced detection go wrong?

Low-energy voiced sounds can be mistaken for unvoiced, while noisy or fricative segments can be mistaken for voiced if the thresholds are not robust. Coarticulation, mixed excitation, DC offsets, quantization noise, and preprocessing all affect the energy and ZCR rules.

Q7: Why do pitch doubling and pitch halving happen?

Pitch doubling happens when the algorithm chooses a submultiple of the true period and returns about $P/2$. Pitch halving happens when the algorithm locks onto about $2P$ instead of P . Both errors are caused by misleading ACF peaks, weak periodicity, waveform symmetry, noisy frames, or poor lag constraints.

Vocoder overview: Q1 to Q5 progression

Stage	What was added	Observed or expected result
Q1	Time-varying LPC synthesis filter only	The vocal-tract filter alone carries some structure, but intelligibility remains limited.
Q2	Frame-energy normalization	Restores amplitude dynamics and rhythm; improves naturalness more than full intelligibility.
Q3	Voiced or unvoiced excitation switching	Better consonant and fricative modeling; less buzzing during unvoiced sounds.
Q4	Time-varying pitch contour	Reduces the fixed-pitch buzz and makes voiced speech more natural.
Q5	Tuned analysis and synthesis, smoothing, residual normalization, aspiration noise	Best overall quality and stability; final audio exported to separate WAV files.

Vocoder Q1 and Q2

Q1: vocal-tract filter only

The first synthesis version uses only the time-varying LPC filter with either pulse-train or noise excitation. This shows that the spectral envelope carries important information, but by itself it does not yield highly intelligible speech.

Q2: adding the energy contour

Energy-contour matching improves intelligibility only moderately, but it clearly restores syllabic rhythm, stress patterns, and amplitude variation over time.

Vocoder Q2: important implementation issue

Failure case discussed in the notebook

With pulse-train excitation and a short update length, some synthesis frames can contain no pitch pulse. Then the output frame energy becomes very small, and naive gain normalization can over-amplify the frame.

Fix proposed in the notebook

Residual-energy normalization and phase-continuous pulse generation are suggested to prevent unstable frame scaling.

Q3: switching excitation using the voiced or unvoiced decision

Using pulse excitation for voiced frames and noise for unvoiced frames improves intelligibility, especially for consonant cues. This avoids using a purely periodic source for fricatives and other noise-like sounds.

Q4: adding time-varying pitch

Introducing a smoothed pitch contour makes voiced segments sound more natural and reduces the fixed-pitch buzz of earlier versions. The final quality depends strongly on the robustness of the pitch estimate.

Vocoder Q3: limitation of binary voicing

Notebook discussion

Real speech is not strictly binary. Mixed excitation occurs in breathy phonation, partially voiced fricatives, aspirated sounds, and transitions.

Implication

A more realistic model would use a continuous voicing measure and generate a weighted mixture of pulse and noise excitation instead of a hard binary switch.

Tuned Q5 system

Tuning steps in the final notebook version

- Smoothed binary voiced or unvoiced mask.
- Median-smoothed pitch track in voiced regions.
- Residual-energy-based excitation normalization.
- Small aspiration-noise component in voiced synthesis.
- Gentle post-correction of frame gain.

Saved output files

male_long_vocoder_q5_alen256_ulen80_M12.wav
female_long_vocoder_q5_alen256_ulen80_M12.wav

